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LDO circuit having power supply rejection function, chip and communication terminal

Abstract

An LDO circuit having a power supply rejection function, a chip, and a communication terminal. The LDO circuit includes a bandgap reference module provided with an intermediate frequency zero adjustment unit, an amplification module provided with an intermediate frequency zero generation unit, and a power output module. The bandgap reference module is connected to the amplification module, and the amplification module is connected to the power output module. For the purpose of power supply rejection at an intermediate frequency, the intermediate frequency zero adjustment unit in the bandgap reference module and the intermediate frequency zero generation unit in the LDO circuit are adjusted in coordination to better optimize the intermediate frequency power supply rejection performance.

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Background/Summary

BACKGROUND

Technical Field

(1) The present invention relates to an LDO circuit having a power supply rejection function, also relates to an integrated circuit chip and a corresponding communication terminal both including the LDO circuit, and relates to the field of radio frequency circuit technologies.

Related Art

(2) With the development of communication technologies, an application environment of a chip in a communication terminal is becoming increasingly more complex. Usually, one communication terminal includes a CPU, a power management chip, a memory chip, a clock chip, a peripheral circuit, a radio frequency chip, and the like, and each part is powered by a same power supply. The CPU, the power management chip, the memory chip, the clock chip, the peripheral circuit, and the like may generate some intermediate frequency signals (100 KHz to 10 MHz). These intermediate frequency signals may enter a voltage bias circuit of the radio frequency chip through the power supply, and then interfere with and deteriorate performance such as radio frequency chip modulation spectrum, switching spectrum, and noise, seriously limiting the development of radio frequency communication technologies.

(3) A low dropout regulator (low dropout regulator, LDO for short) is a voltage bias circuit commonly used in a radio frequency chip. A main function of the regulator is to provide a direct current voltage operating point for the radio frequency chip. To reduce interference of the intermediate frequency signals inside the communication terminal to the radio frequency chip and enhance a capability of the power supply to suppress the intermediate frequency signals, it is necessary to provide an LDO circuit having a high-performance intermediate frequency power supply rejection (Power Supply Rejection, PSR for short) function.

SUMMARY

(4) A primary technical problem to be resolved in the present invention is to provide an LDO circuit having a power supply rejection function.

(5) Another technical problem to be resolved in the present invention is to provide a chip and a communication terminal both including the LDO circuit having a power supply rejection function.

(6) To achieve the above objectives, the following technical solutions are used in the present invention:

(7) According to a first aspect of an embodiment of the present invention, an LDO circuit having a power supply rejection function is provided, including a bandgap reference module provided with an intermediate frequency zero adjustment unit, an amplifier module provided with an intermediate frequency zero generation unit, and a power output module. The bandgap reference module is connected to the amplifier module. The amplifier module is connected to the power output module.

(8) The bandgap reference module generates a reference voltage having a preset temperature coefficient by using a frequency of an intermediate frequency zero adjusted by the intermediate frequency zero adjustment unit and outputs the reference voltage to the amplifier module. The reference voltage is used as a voltage reference of the LDO circuit, and works in coordination with a zero that is generated by the intermediate frequency zero generation unit and whose frequency is an intermediate frequency, to adjust power supply rejection of the LDO circuit at an intermediate frequency.

(9) Preferably, the bandgap reference module includes a starting unit, a PTAT current generation unit, an output unit, and the intermediate frequency zero adjustment unit. An output end of the starting unit and an output end of the intermediate frequency zero adjustment unit are connected to an input end of the PTAT current generation unit. An output end of the PTAT current generation

unit is connected to an input end of the output unit. An output end of the output unit is connected to an input end of the amplifier module.

(10) Preferably, the starting unit includes a first PMOS transistor, a second PMOS transistor, a first resistor, a second resistor, a first NMOS transistor, a second NMOS transistor, a third PMOS transistor, a fourth PMOS transistor, a third NMOS transistor, and a fourth NMOS transistor. A source of the first PMOS transistor and a source of the second PMOS transistor are connected to a power supply voltage. A drain of the second PMOS transistor is connected to one end of the first resistor. The other end of the first resistor is connected to a drain of the first PMOS transistor, a gate of the fourth PMOS transistor, and one end of the second resistor. The other end of the second resistor is connected to a gate and a drain of the first NMOS transistor. A drain of the fourth PMOS transistor is connected to a gate and a drain of the third NMOS transistor and a gate of the fourth NMOS transistor. A drain of the fourth NMOS transistor, a source of the fourth PMOS transistor, and a drain of the third PMOS transistor are connected to each other, and are connected to the PTAT current generation unit with a gate of the second PMOS transistor. A gate of the first PMOS transistor, a gate of the second NMOS transistor, and a gate of the third PMOS transistor each are connected to an external enable signal. The source of the first PMOS transistor, the source of the second PMOS transistor, and a source of the third PMOS transistor are connected to the power supply voltage. A source of the second NMOS transistor, a source of the third NMOS transistor, and a source of the fourth NMOS transistor are grounded.

(11) Preferably, the PTAT current generation unit includes a fifth PMOS transistor, a sixth PMOS transistor, a seventh PMOS transistor, an eighth PMOS transistor, a ninth PMOS transistor, a third resistor, a fourth resistor, a fifth NMOS transistor, a sixth NMOS transistor, a seventh NMOS transistor, an eighth NMOS transistor, a fifth resistor, a first bipolar transistor, and a second bipolar transistor. A gate of the fifth PMOS transistor is connected to an external enable signal. A drain of the fifth PMOS transistor, a gate of the sixth PMOS transistor, a gate of the seventh PMOS transistor, the gate of the second PMOS transistor, a drain of the ninth PMOS transistor, one end of the third resistor, and the output unit are connected to each other. A drain of the sixth PMOS transistor is connected to a source of the eighth PMOS transistor. A drain of the seventh PMOS transistor is connected to a source of the ninth PMOS transistor. A gate of the eighth PMOS transistor, a gate of the ninth PMOS transistor, the drain of the third PMOS transistor, the source of the fourth PMOS transistor, the other end of the third resistor, a drain of the seventh NMOS transistor, and the output unit are connected to each other. A drain of the eighth PMOS transistor is connected to one end of the fourth resistor, a gate of the fifth NMOS transistor, and a gate of the seventh NMOS transistor. The other end of the fourth resistor is connected to a drain of the fifth NMOS transistor, a gate of the sixth NMOS transistor, and a gate of the eighth NMOS transistor. A source of the fifth NMOS transistor is connected to a drain of the sixth NMOS transistor. A source of the seventh NMOS transistor is connected to a drain of the eighth NMOS transistor. A source of the sixth NMOS transistor is connected to an emitter of the first bipolar transistor. A source of the eighth NMOS transistor is connected to an emitter of the second bipolar transistor via the fifth resistor. A source of the fifth PMOS transistor, a source of the sixth PMOS transistor, and a source of the seventh PMOS transistor are connected to the power supply voltage. A base and a collector of the first bipolar transistor and a base and a collector of the second bipolar transistor are all grounded.

(12) Preferably, the output unit includes a tenth PMOS transistor, an eleventh PMOS transistor, a sixth resistor, a third bipolar transistor, a twelfth PMOS transistor, and a thirteenth PMOS transistor. A gate of the tenth PMOS transistor and a gate of the twelfth PMOS transistor are connected to the drain of the ninth PMOS transistor and the intermediate frequency zero adjustment unit. A drain of the tenth PMOS transistor is connected to a source of the eleventh PMOS transistor. A drain of the twelfth PMOS transistor is connected to a source of the thirteenth PMOS transistor. A gate of the eleventh PMOS transistor and a gate of the thirteenth PMOS transistor are connected

to the other end of the third resistor and the intermediate frequency zero adjustment unit. A drain of the eleventh PMOS transistor and one end of the sixth resistor are connected to the amplifier module. The other end of the sixth resistor is connected to an emitter of the third bipolar transistor. A drain of the thirteenth PMOS transistor is connected to the amplifier module. A source of the tenth PMOS transistor and a source of the twelfth PMOS transistor are connected to the power supply voltage. A base and a collector of the third bipolar transistor are grounded.

(13) Preferably, the intermediate frequency zero adjustment unit includes a first capacitor and a second capacitor. One end of the first capacitor and one end of the second capacitor are connected to the power supply voltage. The other end of the first capacitor is connected to the gate of the sixth PMOS transistor, the gate of the seventh PMOS transistor, the gate of the tenth PMOS transistor, and the gate of the twelfth PMOS transistor. The other end of the second capacitor is connected to the gate of the eighth PMOS transistor, the gate of the ninth PMOS transistor, the gate of the eleventh PMOS transistor, and the gate of the thirteenth PMOS transistor.

(14) Preferably, the amplifier module includes a first-stage amplifier unit, a second-stage amplifier unit, and the intermediate frequency zero generation unit configured between the first-stage amplifier unit and the second-stage amplifier unit. The first-stage amplifier unit is connected to the second-stage amplifier unit and the power output module.

(15) Preferably, the intermediate frequency zero generation unit includes a tenth resistor and a fourth capacitor connected in series to ground.

(16) According to a second aspect of an embodiment of the present invention, an integrated circuit chip is provided. The integrated circuit chip includes the LDO circuit having a power supply rejection function.

(17) According to a third aspect of an embodiment of the present invention, a communication terminal is provided. The communication terminal includes the LDO circuit having a power supply rejection function.

(18) In the LDO circuit having a power supply rejection function provided in the present invention, with respect to power supply rejection at an intermediate frequency, the intermediate frequency zero adjustment unit in the bandgap reference module and the intermediate frequency zero generation unit in the LDO circuit are adjusted in coordination to better optimize intermediate frequency power supply rejection performance. When used as a voltage bias circuit of a radio frequency chip, the LDO circuit having the intermediate frequency power supply rejection function can enhance a capability of a radio frequency chip power supply to suppress intermediate frequency signals, thereby improving working performance of the radio frequency chip.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic diagram of an LDO circuit having a power supply rejection function according to an embodiment of the present invention;

(2) FIG. 2 is a schematic diagram of a circuit of a bandgap reference module in an LDO circuit having a power supply rejection function according to an embodiment of the present invention;

(3) FIG. 3 is a schematic diagram of a circuit of an amplifier module and a power output module in an LDO circuit having a power supply rejection function according to an embodiment of the present invention;

(4) FIG. 4 is a comparison diagram of power supply rejection simulation results of a reference voltage output end of a bandgap reference module;

(5) FIG. 5 is a comparison diagram of power supply rejection simulation results of a voltage output end VOUT of an LDO circuit; and

(6) FIG. 6 is an example diagram of a communication terminal using an LDO circuit having a

power supply rejection function.

DETAILED DESCRIPTION

(7) The technical content of the present invention is further described below with reference to the accompanying drawings and the specific embodiments.

(8) To improve an anti-interference capability of a radio frequency chip to intermediate frequency signals in a complex application environment to improve working performance of the radio frequency chip, as shown in FIG. 1, an embodiment of the present invention provides an LDO circuit having a power supply rejection function, including a bandgap reference module **101** provided with an intermediate frequency zero adjustment unit **109**, an amplifier module **102** provided with an intermediate frequency zero generation unit **105**, and a power output module **103**. The bandgap reference module **101** is connected to the amplifier module **102**. The amplifier module **102** is connected to the power output module **103**.

(9) The bandgap reference module **101** generates a reference voltage having a preset temperature coefficient and having an intermediate frequency power supply rejection function by using a frequency of an intermediate frequency zero adjusted by the intermediate frequency zero adjustment unit **109** and outputs the reference voltage to the amplifier module **102**. The reference voltage is used as a voltage reference of the LDO circuit, and works in coordination with a zero that is generated by the intermediate frequency zero generation unit **105** and that is used for changing frequency response of the amplifier module **102** at an intermediate frequency to be an intermediate frequency, to improve intermediate frequency power supply rejection performance of the LDO circuit.

(10) As shown in FIG. 1, the bandgap reference module **101** includes a starting unit **106**, a proportional to absolute temperature (Proportional To Absolute Temperature, PTAT) current generation unit **107**, an output unit **108**, and the intermediate frequency zero adjustment unit **109**. An output end of the starting unit **106** and an output end of the intermediate frequency zero adjustment unit **109** are connected to an input end of the PTAT current generation unit **107**. An output end of the PTAT current generation unit **107** is connected to an input end of the output unit **108**. An output end of the output unit **108** is connected to the amplifier module **102**.

(11) It should be emphasized that in the present invention, an implementation process that the bandgap reference module **101** provides a reference voltage VREF having a preset temperature coefficient (which is generally a zero-temperature coefficient) and having an intermediate frequency power supply rejection function for the amplifier module **102** is mainly described in detail. However, that the bandgap reference module **101** may further provide a bias current for the amplifier module **102** is a well-known existing technology. Details are not described herein.

(12) Specifically, the starting unit **106** is configured to enable the PTAT current generation unit **107** to avoid a degeneracy point where a current is 0, to ensure that the PTAT current generation unit **107** starts normally and is stable. As shown in FIG. 2, the starting unit **106** includes a first PMOS transistor **201**, a second PMOS transistor **202**, a first resistor **203**, a second resistor **204**, a first NMOS transistor **205**, a second NMOS transistor **206**, a third PMOS transistor **207**, a fourth PMOS transistor **208**, a third NMOS transistor **209**, and a fourth NMOS transistor **210**. A connection relationship between each part of the starting unit **106** is as follows: A source of the first PMOS transistor **201** and a source of the second PMOS transistor **202** are connected to a power supply voltage VDD. A drain of the second PMOS transistor **202** is connected to one end of the first resistor **203**. The other end of the first resistor **203** is connected to a drain of the first PMOS transistor **201**, a gate of the fourth PMOS transistor **208**, and one end of the second resistor **204**. The other end of the second resistor **204** is connected to a gate and a drain of the first NMOS transistor **205**. A drain of the fourth PMOS transistor **208** is connected to a gate and a drain of the third NMOS transistor **209** and a gate of the fourth NMOS transistor **210**. A drain of the fourth NMOS transistor **210**, a source of the fourth PMOS transistor **208**, and a drain of the third PMOS transistor **207** are connected to each other, and are connected to a corresponding position of the

PTAT current generation unit **107** with a gate of the second PMOS transistor **202**. A gate of the first PMOS transistor **201**, a gate of the second NMOS transistor **206**, and a gate of the third PMOS transistor **207** each are connected to an external enable signal EN. The source of the first PMOS transistor **201**, the source of the second PMOS transistor **202**, and a source of the third PMOS transistor **207** are connected to the power supply voltage VDD. A source of the second NMOS transistor **206**, a source of the third NMOS transistor **209**, and a source of the fourth NMOS transistor **210** are grounded.

(13) The PTAT current generation unit **107** uses a $R \Delta V_{sub} BE/R$ principle to generate a current proportional to the temperature. As shown in FIG. 2, the PTAT current generation unit **107** includes a fifth PMOS transistor **211**, a sixth PMOS transistor **212**, a seventh PMOS transistor **213**, an eighth PMOS transistor **214**, a ninth PMOS transistor **215**, a third resistor **216**, a fourth resistor **217**, a fifth NMOS transistor **218**, a sixth NMOS transistor **219**, a seventh NMOS transistor **220**, an eighth NMOS transistor **221**, a fifth resistor **222**, a first bipolar transistor **223**, and a second bipolar transistor **224**. A connection relationship between each part of the PTAT current generation unit **107** is as follows: A gate of the fifth PMOS transistor **211** is connected to an external enable signal. A drain of the fifth PMOS transistor **211**, a gate of the sixth PMOS transistor **212**, a gate of the seventh PMOS transistor **213**, the gate of the second PMOS transistor **202**, a drain of the ninth PMOS transistor **215**, one end of the third resistor **216**, and a corresponding position of the output unit **108** are connected to each other. A drain of the sixth PMOS transistor **212** is connected to a source of the eighth PMOS transistor **214**. A drain of the seventh PMOS transistor **213** is connected to a source of the ninth PMOS transistor **215**. A gate of the eighth PMOS transistor **214**, a gate of the ninth PMOS transistor **215**, the drain of the third PMOS transistor **207**, the source of the fourth PMOS transistor **208**, the other end of the third resistor **216**, a drain of the seventh NMOS transistor **220**, and the corresponding position of the output unit **108** are connected to each other. A drain of the eighth PMOS transistor **214** is connected to one end of the fourth resistor **217**, a gate of the fifth NMOS transistor **218**, and a gate of the seventh NMOS transistor **220**. The other end of the fourth resistor **217** is connected to a drain of the fifth NMOS transistor **218**, a gate of the sixth NMOS transistor **219**, and a gate of the eighth NMOS transistor **221**. A source of the fifth NMOS transistor **218** is connected to a drain of the sixth NMOS transistor **219**. A source of the seventh NMOS transistor **220** is connected to a drain of the eighth NMOS transistor **221**. A source of the sixth NMOS transistor **219** is connected to an emitter of the first bipolar transistor **223**. A source of the eighth NMOS transistor **221** is connected to an emitter of the second bipolar transistor **224** via the fifth resistor **222**. A source of the fifth PMOS transistor **211**, a source of the sixth PMOS transistor **212**, and a source of the seventh PMOS transistor **213** are connected to the power supply voltage VDD. A base and a collector of the first bipolar transistor **223** and a base and a collector of the second bipolar transistor **224** are all grounded.

(14) The first PMOS transistor **201**, the second NMOS transistor **206**, the third PMOS transistor **207**, and the fifth PMOS transistor **211** are used as enable control transistors to control on and off of the bandgap reference module **101**. When the enable signal EN is a low level, the first PMOS transistor **201**, the third PMOS transistor **207**, and the fifth PMOS transistor **211** are turned on, the second NMOS transistor **206** is turned off, and the bandgap reference module **101** is off. When the enable signal EN changes from the low level to a high level, the first PMOS transistor **201**, the third PMOS transistor **207**, and the fifth PMOS transistor **211** are turned off, and the second NMOS transistor **206** is turned on. A branch including the second PMOS transistor **202**, the first resistor **203**, the second resistor **204**, the first NMOS transistor **205**, and the second NMOS transistor **206** is on, and a voltage VA at a node A is lowered. The fourth PMOS transistor **208** is turned on and a current is generated. The third NMOS transistor **209** and the fourth NMOS transistor **210** constitute a current mirror. The fourth NMOS transistor **210** is turned on, and a voltage VB at a node B is lower. The eighth PMOS transistor **214** and the ninth PMOS transistor **215** are turned on to enable the PTAT current generation unit **107** to avoid a degeneracy point where a current is 0 and to be

started. After the PTAT current generation unit **107** is stable, the voltage VA at the node A is greater than the voltage VB at the node B, the fourth PMOS transistor **208** is turned off, and the startup of the PTAT current generation unit **107** is completed. The fifth PMOS transistor **211**, the sixth PMOS transistor **212**, the seventh PMOS transistor **213**, the eighth PMOS transistor **214**, the ninth PMOS transistor **215**, the third resistor **216**, the fourth resistor **217**, the fifth NMOS transistor **218**, the sixth NMOS transistor **219**, the seventh NMOS transistor **220**, the eighth NMOS transistor **221** constitute a cascode current mirror bootstrap circuit. Currents in the left and the right branches are equal, and clamping of a node C and a node D is implemented, so that a voltage VC at the node C and a voltage VD at the node D are approximately equal, that is $V_C = V_D$, $V_C = V_{BE_223}$ (which is a base-emitter voltage of the first bipolar transistor **223**), and $V_D = V_{BE_224} + V_{R222}$ (which is a sum of a base-emitter voltage of the second bipolar transistor **224** and a voltage drop across two ends of the fifth resistor **222**). A difference between the base-emitter voltages of the first bipolar transistor **223** and the second bipolar transistor **224** is $\Delta V_{BE} = \ln N \cdot V_T$, where N is a ratio of a quantity of the first bipolar transistor **223** and the second bipolar transistor **224** that are connected in parallel, and V is a thermal voltage between the bipolar transistors. An expression of V_T is

$$(15) V_T = \frac{k}{q} T,$$

where k is a Boltzmann constant, q is electron charge, and T is an absolute temperature, that is,

$$(16) V_{BE} = \ln N \cdot \frac{k}{q} T \cdot \ln N \cdot \frac{k}{q}$$

is a constant. Therefore, the difference ΔV_{BE} between the base-emitter voltages of the first bipolar transistor **223** and the second bipolar transistor **224** is proportional to the absolute temperature T. The voltage drop across two ends of the fifth resistor **222** is $V_{R222} = \Delta V_{BE}$, and each current in the left and the right branches of the cascode current mirror bootstrap circuit is

$$(17) \frac{V_{BE}}{R_{222}} = \frac{\ln N}{R_{222}} \cdot \frac{k}{q} T.$$

(18) The current, that is, a PTAT current, is proportional to the absolute temperature T.

(19) The output unit **108** generates a reference voltage VREF and a reference current IBIAS based on the current that is generated by PTAT current generation unit **107** and that is proportional to the temperature. As shown in FIG. 2, the output unit **108** includes a tenth PMOS transistor **227**, an eleventh PMOS transistor **228**, a sixth resistor **229**, a third bipolar transistor **230**, a twelfth PMOS transistor **231**, and a thirteenth PMOS transistor **232**. A connection relationship between each part of the output unit **108** is as follows: A gate of the tenth PMOS transistor **227** and a gate of the twelfth PMOS transistor **231** are connected to the drain of the ninth PMOS transistor **215** and the intermediate frequency zero adjustment unit **109**. A drain of the tenth PMOS transistor **227** is connected to a source of the eleventh PMOS transistor **228**. A drain of the twelfth PMOS transistor **231** is connected to a source of the thirteenth PMOS transistor **232**. A gate of the eleventh PMOS transistor **228** and a gate of the thirteenth PMOS transistor **232** are connected to the other end of the third resistor **216** and the intermediate frequency zero adjustment unit **109**. A drain of the eleventh PMOS transistor **228** and one end of the sixth resistor **229** are connected to the amplifier module **102**, to provide the reference voltage VREF for the amplifier module **102**. The other end of the sixth resistor **229** is connected to an emitter of the third bipolar transistor **230**. A drain of the thirteenth PMOS transistor **232** is connected to the amplifier module **102**, used for transmitting a reference current IBIAS to the amplifier module **102** to provide the bias current IBIAS for the amplifier module **102**. A source of the tenth PMOS transistor **227** and a source of the twelfth PMOS transistor **231** are connected to the power supply voltage VDD. A base and a collector of the third bipolar transistor **230** are grounded.

(20) Current mirrors are formed respectively by using the tenth PMOS transistor **227** and the eleventh PMOS transistor **228** as well as the twelfth PMOS transistor **231** and the thirteenth PMOS transistor **232**. The PTAT current of the PTAT current generation unit **107** is copied to obtain the reference current IBIAS that is provided for the amplifier module **102** to be used as a bias current. The reference voltage VREF may be expressed as

$$(21) V_{REF} = \left(\frac{\ln N}{R_{222}} * \frac{k}{q} T \right) * R_{229} + V_{BE}$$

$V_{sub.BE}$ is the base-emitter voltage of the third bipolar transistor **230**. The voltage is a voltage with a negative temperature coefficient. Therefore, a ratio of the fifth resistor **222** to the sixth resistor **229** is adjusted, and the reference voltage V_{REF} may become a zero-temperature voltage that is provided for the amplifier module **102** to be used as a voltage reference.

(22) The intermediate frequency zero adjustment unit **109** is configured to adjust a frequency of an intermediate frequency zero of the bandgap reference module **101** to improve the intermediate frequency power supply rejection performance of the bandgap reference module **101**. As shown in FIG. 2, the intermediate frequency zero adjustment unit **109** includes a first capacitor **225** and a second capacitor **226**. One end of the first capacitor **225** and one end of the second capacitor **226** are connected to the power supply voltage V_{DD} . The other end of the first capacitor **225** is connected to the gate of the sixth PMOS transistor **212**, the gate of the seventh PMOS transistor **213**, the gate of the tenth PMOS transistor **227**, and the gate of the twelfth PMOS transistor **231**. The other end of the second capacitor **226** is connected to the gate of the eighth PMOS transistor **214**, the gate of the ninth PMOS transistor **215**, the gate of the eleventh PMOS transistor **228**, and the gate of the thirteenth PMOS transistor **232**.

(23) In the bandgap reference module **101**, the power supply voltage V_{DD} is connected to the other ends of the first capacitor **225** and the second capacitor **226** through each PMOS transistor.

Parasitic capacitances C_{gs} and C_{gd} in the gates exist from a branch to a reference voltage output end. A zero may be generated on the branch. A frequency of the zero is determined by a product of resistance (which is resistance of PMOS transistors) and capacitance on the branch. The frequency of the zero is changed by changing the capacitance, so that power supply rejection of the bandgap reference module **101** at the intermediate frequency is optimized. The frequency of the intermediate frequency zero is determined by both capacitance values of the first capacitor **225** and the second capacitor **226** as well as a ratio between the two capacitors. The capacitance values and the ratio of the first capacitor **225** to the second capacitor **226** may be adjusted with respect to the power supply rejection at the intermediate frequency, and a better optimization effect can be achieved. This is implemented by the following formula.

$$(24) R_{bg} \frac{1}{C_{bg}} = \frac{1}{(r_{ds_227} + r_{ds_228})(((C_{gs1_227} + C_{225}) // C_{gd1_227} // C_{gs2_228}) + C_{226}) // C_{gd2_228}} \quad (1)$$

(25) $R_{sub.bg}$ is resistance on an intermediate frequency zero branch of the bandgap reference module **101**. $C_{sub.bg}$ is the capacitance on the intermediate frequency zero branch of the bandgap reference module **101**. $T_{sub.ds_227}$ is equivalent alternating current resistance between the drain and the source of tenth PMOS transistor **227**. $T_{sub.ds_228}$ is equivalent alternating current resistance between the drain and the source of the eleventh PMOS transistor **228**. $C_{sub.gs1_227}$ is a parasitic capacitance between the gate and the source of the tenth PMOS transistor **227**. $C_{sub.225}$ is the capacitance value of the first capacitor **225**. $C_{sub.ed1_227}$ is a parasitic capacitance between the gate and the drain of the tenth PMOS transistor **227**. $C_{sub.es2_228}$ is a parasitic capacitance between the gate and the source of the eleventh PMOS transistor **228**. $C_{sub.226}$ is the capacitance value of the second capacitor **226**. $C_{sub.gd2_228}$ is a parasitic capacitance between the gate and the drain of the eleventh PMOS transistor **228**.//is a parallel symbol in a circuit.

(26) As shown in FIG. 3, the amplifier module **102** includes a first-stage amplifier unit **110**, a second-stage amplifier unit **111**, and the intermediate frequency zero generation unit **105** configured between the first-stage amplifier unit **110** and the second-stage amplifier unit **111**. The first-stage amplifier unit **110** is connected to the second-stage amplifier unit **111** and the power output module **103**. The power output module **103** includes a power transistor **312**, a sixteenth NMOS transistor **320**, a third capacitor **315**, a seventh resistor **316**, and a feedback resistor network **112** including an eighth resistor R_{f1} and a ninth resistor R_{f2} . A gate of the power transistor **312** is connected to a drain of the sixteenth NMOS transistor **320**, one end of the third capacitor **315**, and

the second-stage amplifier unit **111**. The other end of the third capacitor **315** is connected to one end of the seventh resistor **316**. The other end of the seventh resistor **316** is connected to a drain of the power transistor **312** and one end of the eighth resistor **Rf1**. The other end of the eighth resistor **Rf1** is connected to one end of the ninth resistor **Rf2** and the first-stage amplifier unit **110**. The other end of the ninth resistor **Rf2** is grounded. A source of the power transistor **312** and a source of the sixteenth NMOS transistor **320** is connected to the power supply voltage **VDD**. A gate of the sixteenth NMOS transistor **320** is connected to an external enable signal.

(27) As shown in FIG. 3, the first-stage amplifier unit **110** includes a phase inverter **301**, a ninth NMOS transistor **302**, a tenth NMOS transistor **303**, an eleventh NMOS transistor **304**, a twelfth NMOS transistor **305**, a thirteenth NMOS transistor **306**, a fourteenth NMOS transistor **307**, a fourteenth PMOS transistor **308**, a fifteenth PMOS transistor **309**, and a sixteenth PMOS transistor **319**. A connection relationship between each part of the first-stage amplifier unit **110** is as follows: An input end of the phase inverter **301** is connected to an external enable signal. An output end of the phase inverter **301** is connected to a gate of the twelfth NMOS transistor **305**. A gate of the ninth NMOS transistor **302** and a gate of the sixteenth PMOS transistor **319** each are connected to the external enable signal. A drain of the ninth NMOS transistor **302** is connected to the drain of the thirteenth PMOS transistor **232** in the output unit **108**. A source of the ninth NMOS transistor **302** is connected to a gate and a drain of the tenth NMOS transistor **303**, a gate of the eleventh NMOS transistor **304**, the second-stage amplifier unit **111**, and a drain of the twelfth NMOS transistor **305**. A drain of the eleventh NMOS transistor **304** is connected to a source of the thirteenth NMOS transistor **306** and a source of the fourteenth NMOS transistor **307**. A gate of the thirteenth NMOS transistor **306** is connected to the drain of the eleventh PMOS transistor **228** and one end of the sixth resistor **229** in the output unit **108**. A drain of the thirteenth NMOS transistor **306** is connected to a gate and a drain of the fourteenth PMOS transistor **308**, a gate of the fifteenth PMOS transistor **309**, and a drain of the sixteenth PMOS transistor **319**. A drain of the fifteenth PMOS transistor **309** is connected to a drain of the fourteenth NMOS transistor **307**, the intermediate frequency zero generation unit **105**, and the second-stage amplifier unit **111**. A gate of the fourteenth NMOS transistor **307** is connected to the power output module **103**. A source of the fourteenth PMOS transistor **308**, a source of the fifteenth PMOS transistor **309**, a source of the sixteenth PMOS transistor **319** are connected to the power supply voltage **VDD**. A source of the tenth NMOS transistor **303**, a source of the eleventh NMOS transistor **304**, and a source of the twelfth NMOS transistor **305** are all grounded.

(28) As shown in FIG. 3, the second-stage amplifier unit **111** includes a fifteenth NMOS transistor **310** and a seventeenth PMOS transistor **311**. A gate of the fifteenth NMOS transistor **310** is connected to the gate of the eleventh NMOS transistor **304**. A drain of the fifteenth NMOS transistor **310** is connected to a drain of the seventeenth PMOS transistor **311** and the power output module **103**. A gate of the seventeenth PMOS transistor **311** is connected to the drain of the fourteenth NMOS transistor **307** and the intermediate frequency zero generation unit **105**. A source of the seventeenth PMOS transistor **311** is connected to the power supply voltage **VDD**. A source of the fifteenth NMOS transistor **310** is grounded.

(29) As shown in FIG. 3, the intermediate frequency zero generation unit **105** includes a tenth resistor **317** and a fourth capacitor **318**. One end of the tenth resistor **317** is connected to the drain of the fifteenth PMOS transistor **309** and the gate of the seventeenth PMOS transistor **311**. The other end of the tenth resistor **317** is connected to one end of the fourth capacitor **318**. The other end of the fourth capacitor **318** is grounded.

(30) The first-stage amplifier unit **110**, the second-stage amplifier unit **111**, the power transistor **312**, the eighth resistor **Rf1**, and the ninth resistor **Rf2** form a negative feedback loop, so that an output voltage **VOUT** of the LDO circuit is less affected by changes of a power supply voltage and a load resistance.

(31) Specifically, an enable signal **ENB** is obtained when the enable signal **EN** passes through the

phase inverter **301**. The enable signals EN and ENB control turn-on and turn-off of the ninth NMOS transistor **302**, the twelfth NMOS transistor **305**, the sixteenth PMOS transistor **319**, and the sixteenth NMOS transistor **320** by using enable transistors to turn-on and turn-off of the first-stage amplifier unit **110** and the second-stage amplifier unit **111**. The bandgap reference module **101** provides a bias current for the first-stage amplifier unit **110**. Current mirrors are formed respectively by using the tenth NMOS transistor **303** and the eleventh NMOS transistor **304** and the tenth NMOS transistor **303** and the fifteenth NMOS transistor **310**, and copy the reference current IBIAS to provide the bias current to the first-stage amplifier unit **110** and the second-stage amplifier unit **111**. The bandgap reference module **101** provides a voltage reference for the first-stage amplifier unit **110**. The first-stage amplifier unit **110** is a typical five-transistor amplifier. The thirteenth NMOS transistor **306** and the fourteenth NMOS transistor **307** are input amplifier pair transistors of the first-stage amplifier unit **110**. The fourteenth PMOS transistor **308** and fifteenth PMOS transistor **309** are current mirror loads of the first-stage amplifier unit **110**. The second-stage amplifier unit **111** is a common-source amplifier. The seventeenth PMOS transistor **311** is an amplifier transistor of the second-stage amplifier unit **111**. The third capacitor **315** and the seventh resistor **316** form Miller compensation. A feedback resistor network **112** including the eighth resistor Rf1 and the ninth resistor Rf2, and together with the reference voltage VREF, determine a magnitude of the output voltage of the LDO circuit. The output voltage is

$$(32) VOUT = \frac{Rf1 + Rf2}{Rf2} \cdot VREF.$$

The tenth resistor **317** and the fourth capacitor **318** are connected in series to ground to form the intermediate frequency zero generation unit **105**. A function of the intermediate frequency zero generation unit **105** is to generate a zero whose frequency is an intermediate frequency. The zero may change frequency response of the amplifier module **102** at the intermediate frequency, to improve power supply rejection performance of the intermediate frequency of the LDO circuit. A product of the tenth resistor **317** and the fourth capacitor **318** determines a frequency of the zero. The frequency of the zero is specifically expressed as $1/R_{317}C_{318}$. With respect to power supply rejection at the intermediate frequency, a frequency of an intermediate frequency zero of the bandgap reference module **101** and a frequency of an intermediate frequency zero generated by the intermediate frequency zero generation unit are adjusted in coordination to achieve a better optimization effect.

(33) FIG. 4 shows comparison of power supply rejection (PSR) simulation results of a voltage output end of a bandgap reference module Bandgap. Curve 1 is a power supply rejection simulation result before an intermediate frequency zero adjustment unit is added to the bandgap reference module. Curve 2 is a power supply rejection simulation result after the intermediate frequency zero adjustment unit is added to the bandgap reference module. What is marked in the figure is power supply rejection at a frequency of 300 KHz. According to the simulation result, it can be learned that the power supply rejection at the frequency of 300 KHz is optimized by 11.2 dB after the intermediate frequency zero adjustment unit is added to the bandgap reference module.

(34) FIG. 5 shows comparison of power supply rejection (PSR) simulation results of a voltage output end VOUT of an LDO circuit provided in the present invention. Curve 3 is a power supply rejection simulation result before an intermediate frequency zero generation unit is added to the LDO circuit. Curve 4 is a power supply rejection simulation result after the intermediate frequency zero generation unit is added to the LDO circuit. What is marked in the figure is power supply rejection at a frequency of 300 KHz. According to the simulation result, it can be learned that the power supply rejection at the frequency of 300 KHz is optimized by 11.1 dB after the intermediate frequency zero generation unit is added to the LDO circuit.

(35) In addition, the LDO circuit having a power supply rejection function provided in the present invention may be used in an integrated circuit chip. Details of a specific structure of the LDO circuit having a power supply rejection function in the integrated circuit chip are not described herein.

(36) The LDO circuit having a power supply rejection function may further be used in a communication terminal as an important part of a radio frequency integrated circuit. The communication terminal mentioned here refers to a device that may be used in a mobile environment, supports a variety of communication standards such as GSM, EDGE, TD-SCDMA, TDD-LTE and FDD-LTE, and includes a mobile phone, a laptop, a tablet, an internet of vehicles terminal, and the like. In addition, the technical solutions provided in the present invention are also applicable to a scenario that another radio frequency integrated circuit is used in, such as a communication base station and an intelligent connected vehicle.

(37) As shown in FIG. 6, the communication terminal includes at least a processor and a memory, and may further include a communication component, a sensor component, a power supply component, a multimedia component, and an input/output interface according to actual needs. The memory, the communication component, the sensor component, the power supply component, the multimedia component, and the input/output interface are connected to the processor. The memory may be a static random-access memory (SRAM), an electrically erasable programmable read-only memory (EEPROM), an erasable programmable read-only memory (EPROM), a programmable read-only memory (PROM), a read-only memory (ROM), a magnetic memory, a flash memory, or the like. The processor may be a central processing unit (CPU), a graphic processing unit (GPU), a field-programmable gate array (FPGA), an application-specific integrated circuit (ASIC), a digital signal processing (DSP) chip, or the like. Another communication component, sensor component, power supply component, multimedia component, and the like may be implemented by using a common component. Details are not described herein.

(38) Compared with the conventional technologies, in the LDO circuit having a power supply rejection function provided in the present invention, with respect to power supply rejection at an intermediate frequency, the intermediate frequency zero adjustment unit in the bandgap reference module and the intermediate frequency zero generation unit in the LDO circuit are adjusted in coordination to better optimize intermediate frequency power supply rejection performance. When used as a voltage bias circuit of a radio frequency chip, the LDO circuit having the intermediate frequency power supply rejection function can enhance a capability of a radio frequency chip power supply to suppress intermediate frequency signals, thereby improving working performance of the radio frequency chip.

(39) The foregoing content provides a detailed description of the LDO circuit having a power supply rejection function, the chip, and the communication terminal provided in the present invention. For a person of ordinary skill in the art, any apparent modification made to the present invention without departing from the principle of the present invention falls within the scope of protection of the patent rights of the present invention.

Claims

1. An LDO circuit having a power supply rejection function, comprising a bandgap reference module provided with an intermediate frequency zero adjustment unit, an amplifier module provided with an intermediate frequency zero generation unit, and a power output module, wherein the bandgap reference module is connected to the amplifier module, and the amplifier module is connected to the power output module; wherein the bandgap reference module generates a reference voltage having a preset temperature coefficient by using a frequency of an intermediate frequency zero adjusted by the intermediate frequency zero adjustment unit and outputs the reference voltage to the amplifier module, and the reference voltage is used as a voltage reference of the LDO circuit and works in coordination with a zero that is generated by the intermediate frequency zero generation unit and whose frequency is an intermediate frequency, to adjust power supply rejection of the LDO circuit at an intermediate frequency; wherein the bandgap reference module comprises a starting unit, a PTAT current generation unit, an output unit, and the

intermediate frequency zero adjustment unit, an output end of the starting unit and an output end of the intermediate frequency zero adjustment unit are connected to an input end of the PTAT current generation unit, an output end of the PTAT current generation unit is connected to an input end of the output unit, and an output end of the output unit is connected to an input end of the amplifier module; wherein the starting unit comprises a first PMOS transistor, a second PMOS transistor, a first resistor, a second resistor, a first NMOS transistor, a second NMOS transistor, a third PMOS transistor, a fourth PMOS transistor, a third NMOS transistor, and a fourth NMOS transistor, wherein a source of the first PMOS transistor and a source of the second PMOS transistor are connected to a power supply voltage; a drain of the second PMOS transistor is connected to one end of the first resistor; the other end of the first resistor is connected to a drain of the first PMOS transistor, a gate of the fourth PMOS transistor, and one end of the second resistor; the other end of the second resistor is connected to a gate and a drain of the first NMOS transistor; a drain of the fourth PMOS transistor is connected to a gate and a drain of the third NMOS transistor and a gate of the fourth NMOS transistor; a drain of the fourth NMOS transistor, a source of the fourth PMOS transistor, and a drain of the third PMOS transistor are connected to each other, and are connected to the PTAT current generation unit with a gate of the second PMOS transistor; a gate of the first PMOS transistor, a gate of the second NMOS transistor, and a gate of the third PMOS transistor each are connected to an external enable signal; the source of the first PMOS transistor, the source of the second PMOS transistor, and a source of the third PMOS transistor are connected to the power supply voltage; and a source of the second NMOS transistor, a source of the third NMOS transistor, and a source of the fourth NMOS transistor are grounded; and wherein the PTAT current generation unit comprises a fifth PMOS transistor, a sixth PMOS transistor, a seventh PMOS transistor, an eighth PMOS transistor, a ninth PMOS transistor, a third resistor, a fourth resistor, a fifth NMOS transistor, a sixth NMOS transistor, a seventh NMOS transistor, an eighth NMOS transistor, a fifth resistor, a first bipolar transistor, and a second bipolar transistor, wherein a gate of the fifth PMOS transistor is connected to the external enable signal; a drain of the fifth PMOS transistor, a gate of the sixth PMOS transistor, a gate of the seventh PMOS transistor, the gate of the second PMOS transistor, a drain of the ninth PMOS transistor, one end of the third resistor, and the output unit are connected to each other; a drain of the sixth PMOS transistor is connected to a source of the eighth PMOS transistor; a drain of the seventh PMOS transistor is connected to a source of the ninth PMOS transistor; a gate of the eighth PMOS transistor, a gate of the ninth PMOS transistor, the drain of the third PMOS transistor, the source of the fourth PMOS transistor, the other end of the third resistor, a drain of the seventh NMOS transistor, and the output unit are connected to each other; a drain of the eighth PMOS transistor is connected to one end of the fourth resistor, a gate of the fifth NMOS transistor, and a gate of the seventh NMOS transistor; the other end of the fourth resistor is connected to a drain of the fifth NMOS transistor, a gate of the sixth NMOS transistor, and a gate of the eighth NMOS transistor; a source of the fifth NMOS transistor is connected to a drain of the sixth NMOS transistor; a source of the seventh NMOS transistor is connected to a drain of the eighth NMOS transistor; a source of the sixth NMOS transistor is connected to an emitter of the first bipolar transistor; a source of the eighth NMOS transistor is connected to an emitter of the second bipolar transistor via the fifth resistor; a source of the fifth PMOS transistor, a source of the sixth PMOS transistor, and a source of the seventh PMOS transistor are connected to the power supply voltage; and a base and a collector of the first bipolar transistor and a base and a collector of the second bipolar transistor are all grounded.

2. The LDO circuit having the power supply rejection function according to claim 1, wherein the output unit comprises a tenth PMOS transistor, an eleventh PMOS transistor, a sixth resistor, a third bipolar transistor, a twelfth PMOS transistor, and a thirteenth PMOS transistor, wherein a gate of the tenth PMOS transistor and a gate of the twelfth PMOS transistor are connected to the drain of the ninth PMOS transistor and the intermediate frequency zero adjustment unit, a drain of the tenth PMOS transistor is connected to a source of the eleventh PMOS transistor, a drain of the twelfth

PMOS transistor is connected to a source of the thirteenth PMOS transistor, a gate of the eleventh PMOS transistor and a gate of the thirteenth PMOS transistor are connected to the other end of the third resistor and the intermediate frequency zero adjustment unit, a drain of the eleventh PMOS transistor and one end of the sixth resistor are connected to the amplifier module, the other end of the sixth resistor is connected to an emitter of the third bipolar transistor, a drain of the thirteenth PMOS transistor is connected to the amplifier module, a source of the tenth PMOS transistor and a source of the twelfth PMOS transistor are connected to the power supply voltage, and a base and a collector of the third bipolar transistor are grounded.

3. The LDO circuit having the power supply rejection function according to claim 2, wherein the intermediate frequency zero adjustment unit comprises a first capacitor and a second capacitor; one end of the first capacitor and one end of the second capacitor are connected to the power supply voltage; the other end of the first capacitor is connected to the gate of the sixth PMOS transistor, the gate of the seventh PMOS transistor, the gate of the tenth PMOS transistor, and the gate of the twelfth PMOS transistor; and the other end of the second capacitor is connected to the gate of the eighth PMOS transistor, the gate of the ninth PMOS transistor, the gate of the eleventh PMOS transistor, and the gate of the thirteenth PMOS transistor.

4. The LDO circuit having the power supply rejection function according to claim 1, wherein the amplifier module comprises a first-stage amplifier unit, a second-stage amplifier unit, and the intermediate frequency zero generation unit, and wherein the intermediate frequency zero generation unit is configured between the first-stage amplifier unit and the second-stage amplifier unit, and the first-stage amplifier unit is connected to the second-stage amplifier unit and the power output module.

5. The LDO circuit having the power supply rejection function according to claim 1, wherein the intermediate frequency zero generation unit comprises a tenth resistor and a fourth capacitor connected in series to ground.

6. An integrated circuit chip, comprising the LDO circuit having the power supply rejection function according to claim 1.

7. A communication terminal, comprising the LDO circuit having the power supply rejection function according to claim 1.
